Abstract Algebra

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Lecture 13

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Remark (Commutator)**.** $[x, y] = xyx^{-1}y^{-1}$ gives a measure of how nonabelian a group is.

Theorem 1. (6.1.9) A_n is simple for $n \geq 5$.

Proof. The last theorem proves A_5 is simple. We prove by induction, and assume that A_m is simple for $5 \leq m < n$, and seek to prove that A_n is simple.

Suppose $\{e\} \neq N \lhd A_n$, and try to show that N contains an element σ which leaves an element of $\{1, 2, \ldots, n\}$ fixed. Suppose that no such σ exists. Then there exists distinct elements $a, b, c, d \in \{1, 2, \ldots, n\}$ such that

$$\sigma(a) = b, \sigma(c) = d..$$

Since $n \geq 6$, there is an element $\tau \in A_n$ with

$$\tau = (a, b)(c, d, e, f).$$

for $e, f \notin \{a, b, c, d\}$ with $e \neq f$. By normality of N in A_n, one has $\tau^{-1}\sigma\tau \in N$. Then

$$\begin{aligned} \tau^{-1}\sigma\tau(b) &= \tau^{-1}\sigma(a) = \tau^{-1}(b) = a \\ \tau^{-1}\sigma\tau(d) &= \tau^{-1}\sigma(e) = \tau^{-1}(e) = d. \end{aligned}$$

Then

$$\begin{aligned} \tau^{-1}\sigma\tau\sigma(a) &= \tau^{-1}\sigma\tau(b) = a \\ \tau^{-1}\sigma\tau\sigma(c) &= \tau^{-1}\sigma\tau(d) = d \end{aligned}$$

So $\tau^{-1}\sigma\tau\sigma$ fixes a and moves c, so it is not the identity, and there is an element of N fixing a. We have a contradiction.

We can suppose, by relabeling, that $\sigma \in N \setminus \{\mathrm{id}\}$ and $\sigma(1) = 1$. The elements of A_n which fix 1 form a subgroup of A_n, say $H_1 \leq A_n$, with $H_1 \cong A_{n-1}$. Then $N \cap H_1 \neq \{\mathrm{id}\}$ (since $\sigma \in N \cap H_1$), and $N \cap H_1 \triangleleft H_1$. But H_1 is simple (because A_{n-1} is simple), so $N \cap H_1 = H_1$. Then $H_1 \leq N$.

Let τ be any permutation with $\tau(2) = 1$. Then $\tau^{-1}\sigma\tau(2) = \tau^{-1}\sigma(1) = \tau^{-1}(1) = 2$. So $H_2 := \tau^{-1}H_1\tau$ is a subgroup of A_n that fixes 2. By normality of N in A_n we have $\tau^{-1}H_1\tau \leq N$, and $\tau^{-1}H_1\tau \cong A_{n-1}$. Also,

$$H_1 \cap \tau H_1 \tau \cong A_{n-2}$$

, since it fixes both 1 and 2.

Note, since $H_1 \cap H_2$ fixes both 1 and 2, so $H_1 \cap H_2 \cong A_{n-2}$. Consider the group generated by H_1 and H_2 :

$$\mathrm{gp}(H_1, H_2) = \{g_1, \ldots, g_k : g_i \in H_1 \vee g_i \in H_2\}.$$

Observe that $\mathrm{gp}(H_1, H_2) \leq \mathbb{N}$ since $H_1 \leq N$ and $H_2 \leq N$. Also, the set H_1H_2 is contained in $\mathrm{gp}(H_1, H_2)$ and hence in N.

We compute $|H_1H_2|$. We look at how many representations an element $n \in H_1H_2$ has in the shape $n = h_1h_2$, with $h_1 \in H_1$ and $h_2 \in H_2$.

If $n = g_1g_2$ and $n = h_1h_2$, $g_1, h_1 \in H_1$, $g_2, h_2 \in H_2$, then $g_1g_2 = h_1h_2$. So $H_1 \ni h_1^{-1}g_1 = h_2g_2^{-1} \in H_2$, and hence, for some $t \in H_1 \cap H_2$, we have $h_1^{-1}g_1 = t = h_2g_2^{-1}$.

Then the number of ways that $n \in H_1H_2$ is represented as a product h_1h_2 with $h_1 \in H_1$ and $h_2 \in H_2$ is at most $|H_1 \cap H_2|$. Then

$$|H_1H_2| \geq |H_1||H_2|/|H_1 \cap H_2|.$$

Hence

$$\begin{aligned}
|N| \geq |H_1H_2| &\geq |H_1||H_2|/|H_1 \cap H_2| \\
&= \frac{|A_{n-1}||A_{n-1}|}{|A_{n-2}|} \\
&= \frac{\frac{1}{2}(n-1)!\frac{1}{2}(n-1)!}{\frac{1}{2}(n-2)!} \\
&= \frac{1}{2}(n-1)(n-1)!.
\end{aligned}$$

But $N \triangleleft A_n$, so $|N| \leq \frac{1}{2}n!$. Thus $1 \leq |A_n/N| \leq \frac{\frac{1}{2}n!}{\frac{1}{2}(n-1)(n-1)!} = \frac{n}{n-1} < 2$ when $n > 2$. But now $n \geq 6$, so $|A_n/N| = 1$ and $N = A_n$. Thus A_n is simple for $n \geq 5$. $\square$

Lecture 14: Cauchy's Theorem

Fri 07 Oct 2022 11:10

0.1 Cauchy's Theorem

Definition 1. Let S be a set and $f \in A(S)$. Then the *orbit* of an element $s \in S$ under the action f is $\mathrm{Orb}_f(s) := \{f^i(s) : i \in \mathbb{Z}\}$.

We define an equivalence relation such that

$$s \sim t :\Leftrightarrow t = f^i(s) \text{ for some } i \in \mathbb{Z}.$$

Equivalence classes $[s]$ are then orbits $\mathrm{Orb}_f(s)$.

Lemma 1. (2.8.1) Suppose that $f \in A(S)$ has order p, where p is prime, then for all $s \in S$, we have $|\mathrm{Orb}_f(s)| = 1$ or p.

Proof. Two cases:

1. If $f(s) = s$, then $\mathrm{Orb}_f(s) = \{s\}$, so $|\mathrm{Orb}_f(s)| = 1$.
2. If $f(s) \neq s$, then

$$\mathrm{Orb}_f(s) = \{s, f(s), f^2(s), \dots, f^{p-1}(s)\}.$$

We claim that these elements are all distinct. If not, say $f^i(s) = f^j(s)$, for some $0 \leq i < j \leq p-1$, one has $f^{j-i}(s) = s$. Since also $f^p(s) = s$. But $(p, j-i) = 1$, so there exist $a, b \in \mathbb{Z}$ with $a(j-i) + bp = 1$, and then

$$f(s) = f^{a(j-i)+bp} = f^{bp} f^{(j-i)a}(s) = f^{bp}(s) = s.$$

Therefore $|\mathrm{Orb}_f(s)| = p$ after all.

□

Theorem 2 (Cauchy's Theorem)**.** (2.8.2) Suppose p is prime adn $p \mid |G|$, then G has an element of order p.

Thus G has a subgroup of order p, say $\langle a \rangle \leq G$ with $o(a) = p$.

Proof. Two cases.

1. $p = 2$. If $2 \mid |G|$, if $a^2 = e$ has no solution with $a \neq e$, then for no $a \in G \setminus \{e\}$ does one have $a = a^{-1}$. But for each $b \in G$ there is a distinct $b^{-1} \in G$ with $bb^{-1} = e$. Hence we can partition G into m subsets of shape $\{b, b^{-1}\}$, say together with $\{e\}$. Then $|G| = 2m + 1$, which is a contradiction since $2 \mid |G|$. Thus there exists $a \in G$ with $o(a) = 2$.
2. p prime, $p > 2$, $p \mid |G|$. Let $S = \{(a_1, a_2, \dots, a_p) \in G^p : a_1 \dots a_p = e\}$. Note that if $(a_1, a_2, \dots, a_p) \in S$, then given $a_1, \dots, a_{p-1}$, we have $a_p = (a_1, \dots, a_{p-1})^{-1} \in G$ is uniquely defined, so $|S| = |G|^{p-1}$.

We define

$$\begin{aligned} f : S &\longrightarrow S \\ (a_1, \dots, a_p) &\longmapsto f((a_1, \dots, a_p)) = (a_p, a_1, \dots, a_{p-1}). \end{aligned}$$

Plainly $f \in A(S)$. Note that $f \neq \mathrm{id}$, and $f^p = \mathrm{id}$, so f has order p. Also, one has $f(s) = s$ if and only if $s = (a, \dots, a)$ for some $a \in G$.

Observe that if $f(s) = s$ for precisely m elements $s \in S$, then

$$|S| - m = \text{card}\{s \in S : f(s) \neq s\}.$$

where any element in the set satisfies $|\text{Orb}_f(s)| = p$, so $|S| - m \equiv 0 \pmod{p}$.

Also, $|S| = |G|^{p-1} \equiv 0 \pmod{p}$ since $p \mid |G|$. Thus $m \equiv 0 \pmod{p}$. Since $f((e, \ldots, e)) = (e, \ldots, e)$ and $(\vec{e}.., e) \in S$, $m \geq 1$. Hence $m \geq p > 1$. Hence there exists $a \in G$ with property $(a, \ldots, a) \in S$, whence $a^p = e$. But $o(a)|p$, and $a \neq e$, so $o(a) = p$.

$\square$

Lemma 2. (2.8.3) Let G be a group with $|G| = pq$, where p and q are primes with $p > q$. Whenever $a \in G$ has order p, then $\langle a \rangle \triangleleft G$.

Proof. We claim $A := \langle a \rangle$ is the **only** subgroup of G having order p. Suppose, by the way of deriving a contradiction, that $B \leq G$ and $|B| \neq p$ with $B \neq A$.

Observe that $AB = \{ab : a \in A, b \in B\} \subset G$. Therefore $|AB| \leq pq < p^2$. But if a_1, a_2 are distinct elements of a and b_1, b_2 are distinct elements from B, then $a_1 b_1 = a_2 b_2$ if and only if $a_2^{-1} a_1 = b_2 b_1^{-1}$, so $a_2^{-1} a_1 \in A \cap B$.

But if $B \neq A$, since $A \cap B \leq G$, and A (and B) have prime order, so we must have $A \cap B = \{e\}$. Therefore $a_2^{-1} a_1 = e$, so $a_1 = a_2$ and similarly $b_1 = b_2$. Thus $|AB| = p^2$. But $|AB| < p^2$, a contradiction. Thus A is the only subgroup of G of order p.

Thus if $g \in G$, we have $g^{-1} A g \leq G$ of order p, so $g^{-1} A g = A$. Then $A \triangleleft G$. $\square$

Theorem 3. Suppose G is a group of order pq with p and q primes with $p > q$. Provided that $q \nmid (p-1)$, one has that G is cyclic.

Proof. $\square$